Supplementary material for *Evidence-Monotone Audio Representation: Preventing Provenance Promotion Across Derived Audio Transformations*

Alejandro Urias

S1 Transfer memo: what is prior work and what is new

Every element in the first column is prior work and is cited rather than claimed; every reported number in this paper is new.

Table S1: Relation to the spatial predecessor, cited in the manuscript.

Inherited from the spatial work	New in this work
Evidence record (P, S, A, L); provenance as an atom set with mixed by union	Temporal required-source sets over half-open sample intervals; the captured/generated alphabet and the capture-support channel
Same-channel meet by minimum; scope intersection; lineage union	Requirement (iv): a channel not applicable to *every* required source is withheld, not computed from the rest
Absorbing unverified state	Requirement (iv′): a $\perp$ source also withholds every numeric channel
Non-promotion in the spatial setting	Restated without reference to a modality, as an instance of a known meet-semilattice construction; kernel footprints with declared source-support bounds, validated against stock FFmpeg, tested by differential influence probes and challenged on held-out inputs; footprint monotonicity (Proposition 2)
Conformance-test method (exhaustive enumeration, frozen fixtures, separately implemented oracle)	Signal transparency on decoded PCM; signed C2PA transport with temporal regions; the claim-dilution metric; the strict profile and the consumer verifier

S2 Novelty search and closest-work matrix

Search date First pass 2026-08-19; re-run 2026-09-05 before submission. The re-run is reported below with the first pass, since what a novelty search establishes depends on when it was last attempted.

Databases and interfaces Open-literature web search; arXiv; IACR Cryptology ePrint Archive; ACM Digital Library and LIPIcs listings; the C2PA specification site; OpenSLR.

Query strings (1) `provenance semiring non-amplification derived media operator monotone provenance lattice`; (2) `C2PA content credentials audio provenance derived asset ingredient inheritance research paper 2026`; (3) `"provenance laundering" OR "metadata washing" media provenance attack C2PA`; (4) `audio editing provenance preservation composition proof interval lineage trim concatenate resample formal`; (5) `Denning lattice model secure information flow non-interference applied to media provenance labels 2025 2026`; (6) `audio deepfake detection survey 2026 speech antispoofing ASVspoof 5 generalization`; (7) `audio watermarking soft binding C2PA durable content credentials AudioSeal 2026`.

Outcome The columns *complete-source operator rule* and *C2PA-integrated interval-level enforcement* were empty across every prior row. Because provenance-semiring and information-flow work already yields the non-amplification property abstractly, the theorem is not claimed as new; it is presented as an instance and the contribution is relocated to the executable temporal contract, the transport and the measurements. The full matrix is in `paper/novelty_search.md` in the repository.

Outcome of the re-run The re-run was conducted to falsify the novelty claim rather than to support it, and targeted the three statements this work depends on: that a derived segment must not carry provenance stronger than its weakest required source; that per-operator kernel support is declared and calibrated rather than assumed; and that the resulting interval evidence is carried in a signed C2PA manifest over temporal regions. Within the searched sources, no prior system was found that combines conservative evidence reduction, explicit temporal dependency declarations and a signed C2PA audio transport; the first statement on its own is the meet-semilattice reading of the cited lattice literature and of the spatial predecessor. The searches did return the established algebraic material the paper takes as given, which is the outcome the claim predicts: the meet-semilattice construction and the restriction of semiring provenance to monotone queries are prior art and are cited as such. Two 2026 preprints are the closest adjacent work, one on anonymous provenance for verifiably edited audio and one a security analysis of C2PA and its implementations; both were already cited before the re-run, and neither defines an inheritance rule over media time. C2PA itself records multiple sources as ingredients but specifies no rule constraining what a derived asset may claim relative to them; this work supplies that rule. Nothing found in the re-run changed a claim in this paper or added a reference; a search that surfaces only works already cited is the result we report, not evidence that none was sought. A later journal-context search conducted during target-journal preparation added four FSI:DI papers on forensic representation and tool-output variability for problem framing; none altered the novelty claims or the closest-work assessment above.

S3 Threat-model matrix

Table S2: Threat-model matrix: what each attacker capability meets in the contract, and what the contract does not prevent.

Attacker capability	Contract behaviour	Not prevented
Passes a mixed-origin asset through a benign transformation to obtain a captured-only claim	Output claim is the meet over the complete required source set, so it stays mixed	Nothing, if the attacker also controls the signer

Attacker capability	Contract behaviour	Not prevented
Trims away the boundaries so that only captured material is at the edges	Interior sources remain in D_y; claim stays mixed	A trim that genuinely removes all generated material, which legitimately yields a captured-only claim
Relies on a filtered operator so that a generated sample bleeds into a neighbouring output region without appearing in the nominal range	The declared kernel footprint enlarges D_y; over-approximation is safe by Proposition 2 (footprint monotonicity)	An under-declared footprint, a conformance failure of that implementation; the holdout challenge finds one for the encoder and the stretcher; the strict profile gives every measured declaration whole-asset dependency
Strips the manifest and redistributes	State becomes UNVERIFIED	Detection of the stripping as intentional; re-signing under a fresh, dishonest chain
Modifies the asset after signing	Hard binding fails; state becomes INVALID	Attacks on the cryptographic layer itself
Signs a false assertion with a valid key	Nothing. The contract governs propagation, not truthfulness	Everything: a valid signature proves signing, not truth
Records a loudspeaker playing synthetic speech	A captured claim propagates, correct at the sensor layer	The semantic origin; see the replay limitation in the manuscript

S4 EM assertion schema

The assertion label is `io.github.1d-coda.emaudio.evidence`. The payload is:


```
{
  "schema": "https://github.com/1D-Coda/em-audio/blob/main/docs/em-audio-schema-1.0.md",
  "policy": "complete-source",
  "operator": "trim",
  "operatorParameters": { "start": 4800, "end": 43200 },
  "asset": { "sampleRate": 16000, "sampleCount": 38400 },
  "intervals": [
    {
      "regionOfInterest": {
        "type": "temporal",
        "time": { "type": "npt", "start": "0.000000000", "end": "0.612500000" }
      },
      "samples": { "start": 0, "end": 9800 },
      "provenance": ["C"],
      "state": "CAPTURED",
      "support": { "capture-support": 0.9 },
      "applicability": { "capture-support": ["digital-asset"] },
      "lineage": ["urn:emaudio:librispeech:1272-128104-0000"]
    },
    {
```

```
      "regionOfInterest": {
        "type": "temporal",
        "time": { "type": "npt", "start": "0.612500000", "end": "1.437500000" }
      },
      "samples": { "start": 9800, "end": 23000 },
      "provenance": ["G"],
      "state": "GENERATED",
      "support": { "capture-support": 0.1 },
      "applicability": { "capture-support": ["digital-asset"] },
      "lineage": ["urn:emaudio:espeak-ng:1.52.0:0042"]
    },
    {
      "regionOfInterest": {
        "type": "temporal",
        "time": { "type": "npt", "start": "1.437500000", "end": "2.400000000" }
      },
      "samples": { "start": 23000, "end": 38400 },
      "provenance": ["C"],
      "state": "CAPTURED",
      "support": { "capture-support": 0.9 },
      "applicability": { "capture-support": ["digital-asset"] },
      "lineage": ["urn:emaudio:librispeech:1272-128104-0000"]
    }
  ]
}
```

Normative fields are `provenance` (`null` means unverified and is never an atom), `support`, `applicability` and `lineage`. The `state` field is a presentation label. A channel omitted from `support` is unavailable, which is different from a channel present with a low value. Temporal ranges follow the C2PA `npt` convention with inclusive start and exclusive end; sample-exact bounds are carried alongside because `npt` strings are lossy at sample resolution.

Dependency declaration Schema 1.0 assertions may carry an optional `dependencyDeclaration` object: `version`, `profile` (`declared` or `strict`), `build` (the processing build the footprints were calibrated on), `outputSampleCount`, `sources` (each with its `sampleCount`) and `pieces`, each with its `output` and `sourceRange` spans, `footprint`, `basis` (`analytical`, `measured` or `whole-asset`) and `label`, and `footprintTable`, the module path and SHA-256 digest of the footprint table the declaration was computed from. A consumer rebuilds the map from it and recomputes the intervals (`em_audio/consumer.py`); assertions without it remain valid and are reported as unverifiable. For the trim above, under the strict profile, the declaration is as follows; the digest and the recorded FFmpeg version header are the exact values of the evaluated release.

```
"dependencyDeclaration": {
  "version": "1.0", "profile": "strict",
  "build": "ffmpeg version 9.0.1 Copyright (c) 2000-2026 the FFmpeg developers",
  "footprintTable": {
    "module": "em_audio/operators.py",
    "sha256":
      "f712c10710234527e1a10a06e40943cdce2bda64dae205259b1faa33f26c2f40" },
  "outputSampleCount": 38400,
  "sources": { "clip": { "sampleCount": 48000 } },
  "pieces": [ { "output": { "start": 0, "end": 38400 }, "source": "clip",
                "sourceRange": { "start": 4800, "end": 43200 },
                "footprint": 0, "basis": "analytical", "label": "trim" } ]
```

```
}
```

The `build` field records only the first line of `ffmpeg -version`. That header does not distinguish builds that share a version number, which the independent runs show can differ, and the verifier does not compare it: exact build identity is not enforced in this release. A deployment that needs it should record the executable's digest and the output of `ffmpeg -buildconf` alongside the declaration.

`"clip"` is a local name for a source within the declaration. The consumer binds it to the source's own evidence through the derived asset's `parentOf` ingredient, whose stored manifest carries the source's interval assertion. A composition would bind each name to one `componentOf` ingredient in the same way; the signed workflow evaluated here covers the single-source case only.

The `schema` value is an identifier, not an immutable reference: it is the string stored in every signed fixture and is left unchanged so those signatures remain valid. The authoritative schema for the evaluated results is the copy of `docs/em-audio-schema-1.0.md` archived with release v1.3.0 (the Zenodo record cited in the manuscript's Code availability statement).

S5 Composition manifest wiring

The C2PA-native composition fixtures of manuscript Section 7.9 record each source as a `componentOf` ingredient carrying its full manifest, with a `c2pa.placed` action whose hashed-URI reference the claim generator resolves at signing time. A frozen composition manifest with both ingredients, both resolved `c2pa.placed` references, both source `digitalSourceType` values and the EM aggregate is committed at `fixtures/expected/manifest_composition_componentOf.json`. Wiring pitfalls are listed in `docs/C2PA_TROUBLESHOOTING.md`.

S6 Robustness-arm materials

The neural arm uses Piper `piper-tts` 1.7.0 (MIT, VITS architecture) with the voice `en_US-ljspeech-medium`, whose training data is the LJ Speech Dataset. That dataset states "This dataset is in the public domain in the US (and most likely other countries as well)" and consists of LibriVox recordings by Linda Johnson of non-fiction texts published between 1884 and 1964, all public domain. The `lessac` voices were considered and rejected: their Blizzard 2013 training data carries a restrictive licence. The arm uses that pretrained voice as distributed; no voice is trained or adapted here, and no other speaker's recordings are used.

The model is fetched rather than committed, by `tools/fetch_voice.sh`, which verifies both files against pinned digests. For `en_US-ljspeech-medium.onnx` the SHA-256 is `6f52a751e2349abe7a76735eb09dc1875298c77ea2342ffd2fef79ff81b87f22`, and for `en_US-ljspeech-medium.onnx.json` it is `141d612cc0a95ed7efc1ca936b845c2364967f2e9217c5dbfcf69fc4d6c65860`. Voice output is 22 050 Hz and is resampled to 16 kHz by stock ffmpeg before use.

The noise arm generates pink noise locally, seeded by clip index so the arm is deterministic, and mixes it under the clip at −18 dB. The exact command, as executed:

```
ffmpeg -i <clip> \
  -f lavfi -i "anoisesrc=r=16000:colour=pink:amplitude=0.12:seed=<index>" \
  -filter_complex "[1:a]volume=-18dB[nz];\
                   [0:a][nz]amix=inputs=2:duration=first:normalize=0[out]" \
  -map "[out]" -c:a pcm_s16le <output>
```

The noise is modelled as a second generated-derived source covering the whole clip, so under the contract every captured segment becomes mixed ancestry while the generated segment stays generated, union adding no new atom. Both facts are asserted per interval by the experiment.

S7 Reproduction

```
git clone https://github.com/1D-Coda/em-audio \
  && cd em-audio \
  && git checkout v1.3.0 \
  && python3 tools/fetch_partialspoof.py \
  && python3 tools/fetch_partialedit.py \
  && ./run_all.sh
```

`run_all.sh` fetches the checksum-pinned corpus, generates the test signing credential, runs the named regression tests and experiments A–R, and prints every step it skipped; Experiments Q and R run only when the PartialSpoof and PartialEdit subsets have been fetched, which the two fetch commands do once (each keeps about 70 MB), regenerates every table, macro and figure from the machine-readable results, writes `results/PREFLIGHT.txt`, and exits non-zero if any conformance check fails. The preflight report is the single source of every number in the manuscript; a checker script verifies that no numeric result appears in the manuscript source outside the generated macro file.

What to record The environment: the person or group, the machine, operating system, hardware, runtime and tool versions, and the release tag and commit reproduced. The execution: a clean clone, dependency installation, a full `run_all.sh`, and a comparison of the emitted machine-readable results against those shipped with that tag.

Which outputs must match, and which must not The classification matters more than the checklist, and it is applied by `tools/verify_reproduction.py` rather than by judgement after the fact.

The *deterministic* outputs are those a correct reimplementation must match exactly: conformance counts and pass/fail totals, promotion classifications, lineage sets, interval structure, support-channel behaviour, and the before-and-after-signing identity of every decoded-essence digest. Any difference among them exits non-zero. The digests themselves are not compared across environments: a different decoder build decodes to different bytes, so equality of a digest is defined within one environment, and what must transfer is that signing leaves it unchanged.

The *environment-dependent* outputs are those that should be expected to differ, and whose difference is not a defect: wall-clock timings and their interquartile ranges, the absolute cost figures of Section 7.13 of the manuscript, and the bytes of a freshly signed file, which differ because signing incorporates a timestamp. A reproduction reporting identical timings would be the surprising result, not a reassuring one.

A difference in the first class should be classified rather than absorbed, as an environment difference, a tool-version difference, an ambiguity in the specification of the contract, or a genuine discrepancy, and reported as such whichever it turns out to be.

S8 Independent reproduction: environment and per-operator detail

The run reported in Section 7.14 of the manuscript was performed by D. A. Balderrama-Alvarez (Universidad de Sonora, ORCID 0009-0002-5180-0406), who is not an author. His environment, as his own preflight report records it: a 13th Gen Intel Core i7-1355U with 12 logical processors and 8 GiB, Linux 6.6.87.2 on x86_64, Python 3.14.4, FFmpeg 8.0.1-3ubuntu2, `c2patool` 0.27.15, Node v22.22.1 and eSpeak NG 1.52.0. The reference machine for every other number in the paper is macOS 26.5.2 on arm64 with Python 3.11.15, FFmpeg 9.0.1, `c2patool` 0.27.2 and Node v26.7.0. He ran the package twice on consecutive days. The first run was incomplete: two Python dependencies were missing from the instructions he was given, so the figure and robustness arms did not execute. The second run, on 2026-08-27, is the one released and tabulated here; both runs returned the same values for the two operators whose declarations his build does not satisfy. His unmodified outputs, his preflight report, his full pipeline log and the classified comparison his run produced are released in `results/independent/`.

Two later runs on the released package are summarised in Section 7.14 of the manuscript; their environments are as follows. B. C. Guerra Flores ran it on an Apple M1 under macOS 26.3.1, with FFmpeg 9.0.1 from the same package manager as the reference machine on different hardware, and with a `c2patool` and a Node newer than the reference. Her `verify_reproduction.py` exited zero and every deterministic output matched the package she was given, an earlier release whose suite had 885 828 checks, the 17 differences being timings. D. A. Balderrama-Alvarez ran the released package, tag v1.0.7, as a single unattended pass on Windows 11 with FFmpeg 9.0.1-full_build-www.gyan.dev, a different build of the same FFmpeg version. His verifier reported 69 deterministic differences, all of them in the MP3 kernel measurements of Experiment K and in the time-stretch model deviation, on which the two builds also disagree, 471 against 463 samples, both inside the 2 048-sample margin.

Run history His first run was incomplete because two dependencies were missing from the package instructions, since fixed; the second, reported in the manuscript, gave the same values for both findings. The package he ran, tag v1.0.0, predates the evaluated release v1.3.0; neither finding's value changed between them, which we checked. `tools/verify_reproduction.py` returned `REPRODUCTION INCOMPLETE` with 216 differences, and `run_all.sh` ended in `RUN FAILED` on two packaging defects that touch no reported number. His suite had 885 828 checks, all passing; the released suite has 984 213 because structural checks were added after his run, and the Windows run of tag v1.0.7 reproduces the 984 213 exactly. On his build `aselect` retains 16 384 samples where the reference retains 13 312. The Windows build's time-stretch model deviation is 463 samples against 471 on the reference, both inside the 2 048-sample margin, and the declaration of 2 304 samples contained every macOS and Windows measurement (0 and 0 probes outside).

Cost on his machine His bookkeeping cost is 1.62 ms per audio minute against 1.35 on the reference machine, a difference of 20%; the ratio to the baseline is 3.54 against 3.34, a difference of 5.9%. Across 9 processes in his sitting the absolute figure varied by 13.2% and the ratio by 1.6%. One pair of machines shows the ratio surviving one change of hardware; it does not bound how far it can move.

Table S3 gives the containment result per operator on both machines rather than only the operator that differs, so the selection made in the main text can be checked. 3 of the 4 operators that declare a non-zero footprint reproduce their measured reach exactly across the two builds; the MP3 encoder does not, and its declared footprint does not contain the reach his build measures. The other 3 rows declare no footprint and measure no reach on either machine.

Table S3: Kernel-support containment on the reference machine and on the independent reproducer's machine. *Declared* is the footprint of Table 3 of the manuscript, in source samples; *Reach* is the widest measured influence of a single input sample; *Outside* counts output samples whose dependency falls outside the declared support. The dagger marks the one operator whose reach did not reproduce.

Operator	Declared	Reach (ref.)	Reach (ind.)	Outside (ref.)	Outside (ind.)
retained-run selection	4 096	3 277	3 277	0	0
time stretch 1.10	2 577	1 906	1 906	0	0
transcode to MP3†	2 304	1 555	4 317	0	110
resample 16 to 8 kHz	97	32	32	0	0
amplitude normalisation	0	0	0	0	0
transcode to FLAC	0	0	0	0	0
trim to the middle 80%	0	0	0	0	0

S9 Holdout challenge: per-operator detail

The holdout challenge probes each operator 192 times: 6 signal contexts, 2 of them (white noise and a speech-like voice) unused while the declarations were set, 4 impulse amplitudes from 256 to 12 000, and 8 probe positions disjoint from Experiment K's, all fixed before the run. Its outputs are in `results/machine_readable/K2_footprint_holdout.json`, one row per probe.

The MP3 encoder exceeded its declaration in 4 probes, with a reach of 11 292 source samples, and the time stretcher in 14, with a reach of 15 482; the stretcher's exceedances occur at every amplitude down to 256. Resampling (reach 32), retained-run selection (reach 3 277) and fixed-gain normalisation showed none. With the gain estimated from each input, normalisation reached 15 600 source samples in 84 probes: the whole input, through the gain, which the contract treats as a control dependency rather than represented content. A finite challenge can refute a declaration and cannot establish one, so the absence of an exceedance here does not show containment at every input. Under the strict profile, which gives every measured declaration whole-asset dependency and fits each emitted map to the decoded output length, 0 probes of any content operator fell outside. The promotion demonstrations, one per failing operator, are stored under `promotion_demonstration` in the same file.

S10 Raw rows behind two summary figures

Tables S4 and S5 report the process-level timing measurements and the per-operator calibration summaries. Both are emitted by `tools/make_tables.py` from the same result files the main text reads.

Table S4: The benchmark of Section 7.13 as nine independent processes. The coefficient of variation is the sample standard deviation over the mean, in percent, with Bessel's correction. The absolute columns move with machine state; the last row gives each column's coefficient of variation.

Process	EM (ms/audio-min)	Baseline (ms/audio-min)	Ratio
1	1.4179	0.4083	3.4725
2	1.3753	0.4128	3.3315
3	1.3644	0.4106	3.3233
4	1.3636	0.4125	3.3059
5	1.4081	0.4115	3.4216
6	1.4505	0.4150	3.4954
7	1.3688	0.4101	3.3379
8	1.3738	0.4089	3.3599
9	1.3989	0.4186	3.3422
CV (%)	2.143	0.780	2.046

measured 2026-10-06T10:30:51Z UTC, load average [9.55, 10.42, 11.65]

Table S5: Calibration probes per operator, with the context and source position that produced the widest measured influence. The full per-probe record, all 280 rows, is in `results/machine_readable/CALIBRATION.json`; this table gives the worst case for each operator, which is the value the declaration has to cover. The 7 rows are the single-source configurations: concatenation assembles the corpus rather than transforming it, and overlay takes several sources at once.

Operator	Probes	Max reach	Worst context	At sample	Declared
retained-run selection	40	3 277	tone	6 569	4 096
time stretch 1.10	40	1 906	tone	6 537	2 577
transcode to MP3	40	1 555	dense	8 192	2 304
resample 16 to 8 kHz	40	32	tone	1 638	97
trim to the middle 80%	40	0	tone	64	0
transcode to FLAC	40	0	tone	64	0
amplitude normalisation	40	0	tone	64	0